

Simulation with Envi-guide

1

ENVI-guide V5.5.1

Start the Envi-guide. Select the folder of your project.

Simulation Date and Time

Start Date (YYY MM DD): 2022 07 25

Start Time (HH:MM): 5 : 0

Total Simulation Time (h): 24

Simulation name

Full name of simulation task: AltstettenLinden
This is used to identify your simulation and to generate labels

Short name for file names: AltstettenLinden
Define the root name for your simulation files.
ENVI-met will add some information to this name, so keep it simple but unique

Folder for model outputs: C:\EnviMetProjects\Altstetten\Altstetten\
If left empty, the outputs will be written next to the SIMX file

Model area

Load model area (INX) file: inx_file.INX

Set the "folder for model outputs" and "load model area (INX) file".
For outputs, create a new folder in your project folder, e.g., C:\EnviMetProjects\Altstetten\Altstetten-baseline-output.

CPU Cores (Parallel Computing)

Number of CPU cores to be used
☒ Multi Core (parallel) ☐ Single Core

Decide whether the simulation should utilize all available CPU cores. While this typically increases speed, other applications might not run as smoothly.

(Parallel Computing is not supported in LITE and STUDENT)

2

ENVI-guide V5.5.1

Select Meteorology section.

Select Meteorology

Meteorological Boundary Conditions

Use Simple Forcing

Use Full Forcing

Use Other LBC

Select "Simple forcing".
Your selection should be based on the detail and variety of climate data you have. If you lack observational data and are testing hypothetical urban scenarios, you may also opt for "Other LBC".

ENVI-met simulations need some information about the surrounding meteorology. The more information the model is provided, the more exact a climate scenario can be modelled.

Two different meteorological boundary conditions can be used to provide information to the model:

- Simple Forced [recommended, if only information about Air Temperature and Humidity are present]
- Full Forced [recommended, represents the highest precision]

The best results can be expected by using the full forcing option, however more information about the surrounding meteorology is needed to run the model.

Older options (see other LBC option) which, due to their simplicity, are not recommended are:

- Open

Simulation with Envi-guide

3

ENVI-guide V5.5.1

ENVI-guide

Help

New SIMX

Load SIMX

Save SIMX

Save SIMX as

New / Open

Save Configuration

Sections

Click on Sections to adjust parameters

Mandatory Sections

General Settings

Meteorology

Use Simple Forcing

Optional Sections

Adjust the values for the T and rH based on your observations or EPW file for the simulation day

You have two options to set diurnal cycle based on air temperature and humidity:

1- enter observations for each hour.

2- generate it through automatic linear interpolation.

Set constant wind and radiation conditions.

Add Optional Section

Remove Optional Section

Meteorology: Simple Forcing

Set 24-hour cycle of Air Temperature and Humidity

Air Temperature and Humidity

Temp (°C)

Hour

75

65

55

45

30

29

28

27

26

25

24

0

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

21

22

23

Air Temperature

rel. Humidity

Manually adjust values

Time	T	rH
00:00	25.4	58
01:00	24.6	66
02:00	23.7	70
03:00	23.3	77
04:00	22.8	76
05:00	22.5	77
06:00	22.7	73
07:00	23.9	67
08:00	25.1	61
09:00	26.3	60
10:00	27.4	53
11:00	28.4	49
12:00	29.3	50
13:00	29.9	46
14:00	30.4	43
15:00	30.7	44
16:00	30.7	44
17:00	30.5	47
18:00	30	49
19:00	29.2	56
20:00	28.2	58
21:00	27.3	64
22:00	26.4	66
23:00	25.4	70

Create 24-hour cycle by automatic linear interpolation

Time of Max Air Temperature: 16

Min Air Temperature: 21 °C

Max Air Temperature: 28 °C

Time of Min Air Temperature: 5

Time of Min Rel Humidity: 8

Min relative Humidity: 45 %

Max relative Humidity: 75 %

Time of Max Rel Humidity: 4

Update

Humidity in 2500 m

Specific humidity in 2500 m (g/kg): 13.48

Set Boundary Conditions for Wind and Radiation

Wind and Radiation

Windspeed

Constant windspeed at inflow border (m/s): 0.87

Wind direction

Constant wind direction at inflow (°): 224.38

Roughness Length

Microscale roughness length of surface (m): 0.010

Low clouds

Cloud cover of low clouds (0-8): 0

Medium clouds

Cloud cover of medium clouds (0-8): 0

High clouds

Cloud cover of high clouds (0-8): 0

4

ENVI-guide V5.5.1

ENVI-guide

Help

New SIMX

Load SIMX

Save SIMX

Save SIMX as

New / Open

Save Configuration

Sections

Click on Sections to adjust parameters

Mandatory Sections

General Settings

Meteorology

Use Full Forcing

Optional Sections

In case you want to run your simulation with "Full Forcing"...

Use Simple Forcing

Use Full Forcing

Use Other LBC

Add Optional Section

Remove Optional Section

Meteorology: Full Forcing

Provide Meteorological Data

Meteorology for Full Forcing (either select an existing FOX-File or create a new one)

Forcing files in project folder:

FOX-Files

Click on "Add FOX File"

Add FOX File

Refresh folder

Forcing file to be used in simulation:

FOX-File

No Forcing File loaded yet

Which data shall be forced? Please provide alternate data instead

Do you want to force wind?

Do you want to force air temperature?

Do you want to force radiation/clouds?

Do you want to force rel. humidity?

Do you want to force precipitation?

Adjust wind inflow updates

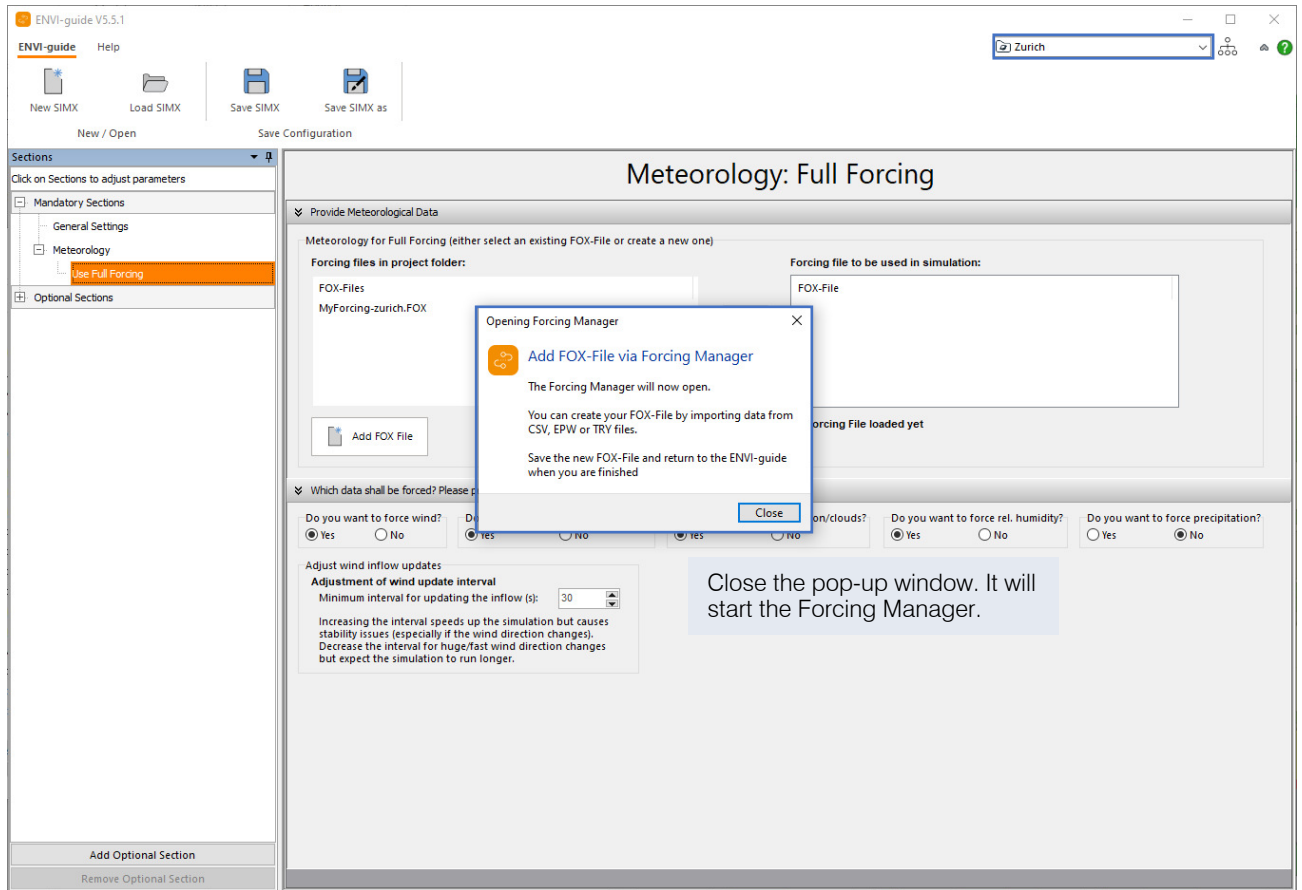
Adjustment of wind update interval

Minimum interval for updating the inflow (s): 30

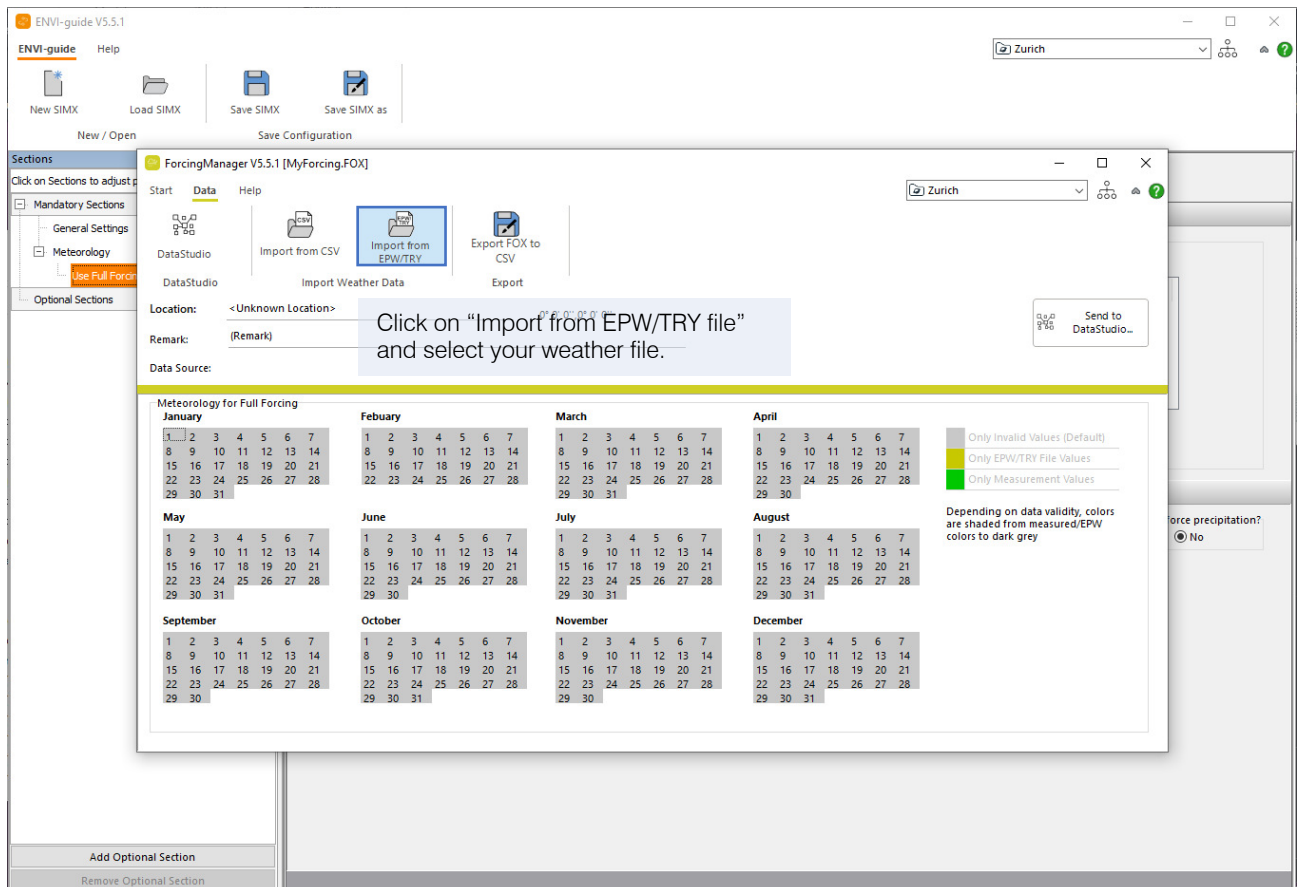
Increasing the interval speeds up the simulation but causes stability issues (especially if the wind direction changes). Decrease the interval for huge/fast wind direction changes but expect the simulation to run longer.

Simulation with Envi-guide

5



6



Simulation with Envi-guide

7

ENVI-guide V5.5.1

ForcingManager V5.5.1 [MyForcing.FOX]

Rename the location if necessary. Choose 'Save as FOX-file' to store it in your project folder. Dates for which you have data should turn green.

Location: Zürich 8° 32' 20", 47° 22' 11"

Remark:

Data Source: EPW

Meteorology for Full Forcing

January, February, March, April, May, June, July, August, September, October, November, December

Values for: 18. Juli

air temperature (°C), specific humidity (g/kg), wind speed (m/s), precipitation (mm)

Depending on data validity, colors are shaded from measured/EPW colors to dark grey

Send to DataStudio...

8

ENVI-guide V5.5.1

Meteorology: Full Forcing

Provide Meteorological Data

Meteorology for Full Forcing (either select an existing FOX-File or create a new one)

Forcing files in project folder:

FOX-Files: MyForcing-zurich.FOX

Forcing file to be used in simulation:

FOX-File: MYFORCING-ZURICH.FOX

Forcing Data loaded successfully and covers Simulation Time!

Which data shall be forced? Please provide alternate data instead

Do you want to force wind? Yes No

Do you want to force air temperature? Yes No

Do you want to force radiation/clouds? Yes No

Do you want to force rel. humidity? Yes No

Do you want to force precipitation? Yes No

Adjust wind inflow updates

Adjustment of wind update interval

Minimum interval for updating the inflow (s): 30

Increasing the interval speeds up the simulation but causes stability issues (especially if the wind direction changes). Decrease the interval for huge/fast wind direction changes but expect the simulation to run longer.

Close the Forcing Manager and then refresh the folder. You will find the FOX Manager file you created in this folder. Select this file as the forcing file for use in the simulation.

Simulation with Envi-guide

9

ENVI-guide V5.5.1

ENVI-guide Help

New SIMX Load SIMX Save SIMX Save SIMX as

New / Open Save Configuration

Sections

Click on Sections to adjust parameters

Mandatory Sections

General Settings

Meteorology

Use Full Forcing

Optional Sections

Meteorology: Full Forcing

Provide Meteorological Data

Meteorology for Full Forcing (either select an existing FOX-File or create a new one)

Forcing files in project folder:

FOX-Files

MyForcing-zurich.FOX

Add FOX File Refresh folder

Forcing file to be used in simulation:

FOX-File

MYFORCING-ZURICH.FOX

Forcing Data loaded successfully and covers Simulation Time!

Which data shall be forced? Please provide alternate data instead

Do you want to force wind? ☐ Yes ☒ No

Do you want to force air temperature? ☒ Yes ☐ No

Do you want to force radiation/clouds? ☐ Yes ☒ No

Do you want to force rel. humidity? ☒ Yes ☐ No

Do you want to force precipitation? ☐ Yes ☒ No

Alternate wind speed and direction (not full forced)

Windspeed

Constant windspeed at inflow border (m/s): 2.00

Wind direction

Constant wind direction at inflow (°): 135.00

Roughness Length

Microscale roughness length of surface (m): 0.010

Alternate cloud cover (not full forced)

Low clouds

Cloud cover of low clouds (0-8): 0

Medium clouds

Cloud cover of medium clouds (0-8): 0

High clouds

Cloud cover of high clouds (0-8): 0

Select the parameters you wish to force. In this scenario, the case is very similar to 'Simple Forcing'.

Add Optional Section

Remove Optional Section

10

ENVI-guide V5.5.1

ENVI-guide Help

New SIMX Load SIMX Save SIMX Save SIMX as

New / Open Save Configuration

Sections

Click on Sections to adjust parameters

Mandatory Sections

General Settings

Meteorology

Use Full Forcing

Optional Sections

Add Optional Sections to SIMX

Soil Section

Radiation Section

Building Section

Pollutant Section

Plant Section

Timing Section

Output Section

Expert Section

Click on 'Add Optional Section' if you wish to specify additional settings for various categories.

Add Optional Section

Remove Optional Section

Simulation with Envi-guide

11

ENVI-guide V5.5.1

ENVI-guide Help

New SIMX

Load SIMX

Save SIMX

Save SIMX as

New / Open

Save Configuration

Sections

Click on Sections to adjust parameters

Mandatory Sections

General Settings

Meteorology

Use Full Forcing

Optional Sections

Soil Section

Add Optional Section

Remove Optional Section

Soil Conditions

Initial Soil Conditions

Soil Layer	Soil Humidity (%)	Initial temperature (°C)
Upper layer (0-20 cm)	65.00	20.00
Middle layer (20-50 cm)	70.00	20.00
Deep layer (50- 200 cm)	75.00	19.00
Bedrock layer (below 200 cm)	75.00	18.00

Set the soil temperature and humidity in the different layers under ground. Soil humidity is given in Usable Field Capacity (Usable Field Capacity = Field Capacity - Wilting Point). If you want to e.g. simulate a day in a long lasting dry period, you should modify the values accordingly.

You have the option to adjust settings pertaining to the water content and the initial temperature of various soil layers.

12

ENVI-guide V5.5.1

ENVI-guide Help

New SIMX

Load SIMX

Save SIMX

Save SIMX as

New / Open

Save Configuration

Sections

Click on Sections to adjust parameters

Mandatory Sections

General Settings

Meteorology

Use Full Forcing

Optional Sections

Soil Section

Building Section

Add Optional Section

Remove Optional Section

Building Settings

Building Indoor Temperature

Building indoor temperature [°C]: 24.50

Initial building surface temperature [°C]: 20.00

Keep the Building Indoor Temperature constant?
☐ Yes ☒ No

Set the initial temperature inside of your buildings. If you choose to keep the building temperature constant, the building temperature will not be influenced by the outside microclimate.

You can also set the building surface temperature and the indoor temperature.
Keeping the indoor temperature constant implies that you are climatizing the indoor environment, ensuring that the outside microclimate does not influence the indoor conditions. These settings significantly impact the Building Energy Simulation (BES) in Envi-met.

Simulation with Envi-guide

13

ENVI-guide V5.5.1

ENVI-guide Help

New SIMX

Load SIMX

Save SIMX

Save SIMX as

New / Open

Save Configuration

Sections

Click on Sections to adjust parameters

Mandatory Sections

General Settings

Meteorology

Use Full Forcing

Optional Sections

Soil Section

Building Section

Plant Section

Add Optional Section

Remove Optional Section

Plant Settings

General Plant Settings

Leaf Transmittance

Use old calculations

Use user defined values

Use Tree Calendar

Yes

No

CO2 Background Concentration

The default value of the CO2 background concentration in the atmosphere is set to 400 ppm. To compare new simulation results with old results prior ENVI-met v4.4, you might want to set it back to 350 ppm. For future scenarios, you may want to increase the value.

In general, CO2 concentration in the simulation is used for photosynthesis processes of plants. This value is not connected to the User Pollutant that can be defined as CO2. If you want to increase that background concentration, you will then have to increase the background concentration of the User Pollutant in the Pollutant Section.

CO2 background level (ppm):

400

(default value 350 ppm prior to ENVI-met v4.4)

You have the option to adjust general plant settings and CO2 concentration, if necessary. Typically, we don't change these settings.

14

ENVI-guide V5.5.1

ENVI-guide Help

New SIMX

Load SIMX

Save SIMX

Save SIMX as

New / Open

Save Configuration

Sections

Click on Sections to adjust parameters

Mandatory Sections

General Settings

Meteorology

Use Full Forcing

Optional Sections

Soil Section

Building Section

Plant Section

Radiation Section

Add Optional Section

Remove Optional Section

Radiation Settings

Ray Tracing and IVS Height Cap

Raytracing

Raytracing precision

Finer resolution (slower)

Lower resolution (faster)

Use height cap for adjusting raytracing (and IVS) precision?

Yes

No

Do you want to use a height cap above which the raytracing precision / IVS resolution changes? [What is meant by that?](#)

10

Height cap in meters above ground below which higher precision is used.

Indexed View Sphere (IVS)

Use IVS module?

Yes

No

Resolution for height segment angles:

Lowest (45°)

Resolution for azimuthal segment angles:

Lowest (45°)

The Indexed View Sphere (IVS) algorithm enables a higher level of detail regarding the calculation of reflected shortwave and longwave radiation fluxes by taking into account multiple interactions between objects (vegetation, buildings, surfaces). Segment angle sizes refer to the number of segments to be calculated for each view sphere. As angle values decrease, accuracy of received radiation as well as number of segments increase. The higher level of detail massively increases RAM demand.

Mean Radiant Temperature Calculation Method and Minimum Longwave Radiation Cap

Mean Radiant Temperature (MRT)

MRT calculation method

Two-directional approach (ENVI-met V4.X)

Common six-directional approach

Please choose whether simulation shall feature the old two-directional ENVI-met or the common six-directional method (new default).

Human projection factor:

VDI 2008 (RayMan, SURM)

The projection factor estimates the surface of a human hit by radiation. Different calculation methods can be selected.

Advanced Canopy Radiation Module and Solar Adjustment Factor

Advanced Canopy Radiation Transfer Module (ACRT)

Use ACRT module?

Yes

No

Advanced Canopy Radiation Transfer describes a new approach where direct shortwave radiation is partly attenuated and partly scattered within plants according to their LAD. When scattered, diffuse shortwave radiation is created. With this ACRT module, plant effects are simulated more accurately.

Update Interval for View Factor Calculation for the ACRT module in days (accounting for seasonal leaf area changes cf. tree calendar)

10

Solar Adjustment Factor

Adjustment factor for solar radiation:

1.00

Preview radiation...

Modify the irradiation of your model area [W] by increasing or decreasing the solar adjustment factor. Please note that this is not recommend in general, since the radiation values are calculated correctly based on the sun's position which is derived by date/time and model area location. If you want to adjust the irradiation, please use cloud cover or force the radiation values hourly by using the FullForcing method. This option is only available due to legacy reasons.

In this section, you can access the detailed settings for radiation. Typically, these settings are left unchanged, as they are already configured to a moderately high resolution.

A / S

Architecture and Building Systems

Prof. Dr. Arno Schlueter
Institute of Technology in Architecture (ITA)
ETH Zurich

prepared by
Ayca Duran

7

Simulation with Envi-guide

15

ENVI-guide V5.5.1

ENVI-guide Help

New SIMX

Load SIMX

Save SIMX

Save SIMX as

New / Open Save Configuration

Sections

Click on Sections to adjust parameters

Mandatory Sections

General Settings

Meteorology

Use Full Forcing

Optional Sections

Soil Section

Building Section

Plant Section

Radiation Section

Pollutant Section

Add Optional Section

Remove Optional Section

Pollutants / Sources Settings

General Pollutant Simulation Settings

The pollutant emission rates are taken from the emission profiles defined in the Database Manager. In Single pollutant mode, only the user-defined pollutant is used. In MultiPollutant Mode, all other emission profiles such as NO, O3 etc. are used as well.

Pollutant Operation Mode

Single Pollutant

Multi Pollutant

Using the additional Active Chemistry module requires Multi Pollutants mode. Active chemistry supports reaction terms of NO, O3 and NO2.

Active Chemistry (NO-O3-NO2)

Dispersion only

Dispersion and Active Chemistry

(Active Chemistry is not supported in LTE)

The update interval for emission rates can be adjusted in the Timing Section.

Background Concentration

Set the background concentration of different pollutants if needed. Similar to the CO2 background concentration in the atmosphere (~400 ppm) that can be changed in the Plants Section, there might be a specific pollutant background concentration for your simulated area. Default values are set to 0 in order to assume a clean atmosphere for the simulation start.

NO background level (µg/m3): 0.00

NO2 background level (µg/m3): 0.00

Ozone background level (µg/m3): 0.00

PM 10 background level (µg/m3): 0.00

PM 2.5 background level (µg/m3): 0.00

User specified pollutant (µg/m3): 0.00

User Pollutant / User Emission Profile Settings

Name of User Pollutant Source: My Pollutant

Enter a name to identify the pollutant in the output files.

You need to specify a user-defined pollutant in Single pollutant mode!

Chemical species or type of pollutant: <None>

Should you have modeled pollutant sources, such as traffic, you can adjust the related settings in this section.

16

ENVI-guide V5.5.1

ENVI-guide Help

New SIMX

Load SIMX

Save SIMX

Save SIMX as

New / Open Save Configuration

Sections

Click on Sections to adjust parameters

Mandatory Sections

General Settings

Meteorology

Use Full Forcing

Optional Sections

Soil Section

Building Section

Plant Section

Radiation Section

Pollutant Section

Timing Section

Add Optional Section

Remove Optional Section

Simulation Timing Settings

Update Timing

Plant Processes (s): 600

Flow Field (s): 900

Surface Data (s): 30

Emission Rates (s): 600

Radiation / Shadows (s): 600

Define the time interval between updating different processes like shadow casting or surface data. Using an update interval of 0 s means continuous calculation! This will slow down the calculation time dramatically!

Dynamic Time Step Management

ATTENTION: This section provides the possibility to decrease timesteps in order to avoid stability problems. It is strongly recommend to NOT increase timesteps!

Time step t0 (s): 2.00

Time step t1 (s): 2.00

Time step t2 (s): 1.00

Solar angle for switching t0 to t1 (deg): 40.00

Solar angle for switching t1 to t2 (deg): 50.00

Set the time interval in seconds between the different time steps. If the model becomes unstable, you might want to adjust the time steps of the solar height switch points

In this section, you can access the settings for updating the model at specific time intervals. These settings are typically left unchanged.

Simulation with Envi-guide

17

ENVI-guide V5.5.1

ENVI-guide Help

New SIMX

Load SIMX

Save SIMX

Save SIMX as

New / Open

Save Configuration

Sections

Click on Sections to adjust parameters

Mandatory Sections

General Settings

Meteorology

Use Full Forcing

Optional Sections

Soil Section

Building Section

Plant Section

Radiation Section

Pollutant Section

Timing Section

Output Section

Add Optional Section

Remove Optional Section

Output Settings

Output Files

☒ Building Data (Facade Temperature, Roof Temperature, etc.)

☒ Radiation Data (Shortwave Radiation, Longwave Radiation, etc.)

☒ Soil Data (Soil Wetness, Soil Temperature, etc.)

☒ Vegetation Data (Leaf Temperature, Transpiration Flux, etc.)

Decide which output files should be saved. Please be aware, that by de-selecting outputs, these will not be saved in the simulation.

Output Interval: Receptors and buildings (min):

5

Output Interval: All other files (min):

60

Decide in which output interval you want to save the receptor and building data and all other files.

NetCDF Output Files

Write supplementary NetCDF output

☐ Yes

☒ No

Check if NetCDF files shall be written additional to the ENVI-met default outputs. All ENVI-met output variables (out of the folders atmosphere, buildings, radiation etc.) will be stored in one file (unless the small file option is selected at the bottom of this NetCDF settings section).
NetCDF is a universal file format for multidimensional datasets, commonly used for climate models. Multiple libraries and software packages allow a broad range of independent analysis possibilities.
Please install the netCDF-C library "netCDF-4 64-bit (Windows)" to enable NetCDF file writing (download can be found on the official netCDF Website: <https://downloads.unidata.ucar.edu/netcdf/>).

NetCDF: One or multiple output files

☒ Data of every output interval is stored in one single NetCDF file.

☐ A NetCDF file is created for every output interval.

Large or small NetCDF output file

☒ All existing output variables are stored in the file.

☐ Only most relevant variables (like Objects, Radiation, T, Q, MRT) are stored in the file to save disk space.

You have the option to adjust the time interval for saving outputs. Additionally, you can save your results as a NetCDF file. Typically, these settings are not modified.

Include Nesting grids in output files

☐ Yes

☒ No

Decide whether (possibly added) nestings grids shall be included in the results.

Attention: Nesting grids are always written in NetCDF outputs.

18

ENVI-guide V5.5.1

ENVI-guide Help

New SIMX

Load SIMX

Save SIMX

Save SIMX as

New / Open

Save Configuration

Sections

Click on Sections to adjust parameters

Mandatory Sections

General Settings

Meteorology

Use Full Forcing

Optional Sections

Soil Section

Building Section

Plant Section

Radiation Section

Pollutant Section

Timing Section

Output Section

Expert Section

Add Optional Section

Remove Optional Section

Specific Expert Settings

SOR mode

SOR

☐ Use old pressure field

☒ Use new pressure field

The new pressure field is calculated via the red-black-tree algorithm, which allows a parallel computation of the pressure field.

Turbulence Model (TKE)

Choose TKE model:

Bruse/ENVI-met 2017

The different models mainly vary in the calculation of the mechanical production term P_m , the deformation of the wind field, and the rotation of the wind field. For further information, you can read the corresponding papers.

Facades

Method

☐ MO (default prior to v4.4)

☒ DIN 6946 (new default)

Set how the wind resistance at facades is calculated. DIN 6946 includes a higher z_0 value and is the new default value.

Simulation Threading

Threading

☒ Run simulation in main thread

☐ Run simulation in its own thread

When running in its own thread, the application will stay responsive. However, multiple simulations on the same PC might cause instabilities and detailed error messages are not supported.

Averaged Inflow

Use Average Inflow

☐ Yes

☒ No

If activated, the reference surface temperature / humidity will be calculated accounting only inflow cells. Otherwise, it will be calculated using the average values of each grid cell.

In this section, you can access expert settings related to the calculation methods used in the simulation. Typically, these settings are left as default.